Fires, Lights, and Smog: Reading Bolivia's Recession from Space*

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Abstract

This paper constructs a monthly coincident indicator of Bolivian real activity from four independent satellite streams: VIIRS Day/Night Band nighttime radiance over eleven urban buffers, VIIRS Nightfire radiant heat over the six major Chaco gas fields, Sentinel-5P TROPOMI tropospheric nitrogen-dioxide columns over the La Paz–El Alto and Santa Cruz metropolitan areas, and Sentinel-2 NDVI over five cropland zones. The four streams together cover approximately 78 percent of GDP, leaving mainly non-tradeable services as an explicit blind spot. The streams are combined through a two-factor mixed-frequency dynamic factor model separating urban from extractive activity, with an agricultural factor conditioning on NDVI, and anchored to quarterly GDP via the Mariano and Murasawa (2003) aggregation identity. The indicator arrives at monthly frequency with a two-month publication lag, against a two- to three-month lag on INE quarterly GDP and a March 2028 expected date for twenty-four vintages of INE's new experimental IGAE. I apply the indicator to Bolivia's 2025 to 2026 recession and demonstrate three manipulation-detection tests built around single-country sources of identification: sectoral triangulation across VIIRS Nightfire, YPFB-reported gas production, and INE hydrocarbon value added; a November 2025 INE-leadership discontinuity regression; and external-forecaster residual falsification. **[PENDING: Headline: trough date, depth, and sectoral decomposition, inserted after the pipeline completes a full vintage.]**

JEL: C32, C38, E01, E32, O11, O54, Q15, Q35, Q43.
Keywords: nighttime lights, VIIRS, TROPOMI, NDVI, dynamic factor model, nowcasting, coincident indicator, GDP manipulation, Bolivia, natural gas, recession dating.

*All opinions are the author's own and do not reflect the views of any affiliated institution. All remaining errors are mine. Contact: `wernerhl@gmail.com`. Replication materials, satellite processing pipeline, and frozen dataset will be made available at a Zenodo DOI upon publication.

1 Introduction

Bolivia entered a recession of contested depth in 2025. The national statistics office (INE) reported a −1.58 percent contraction for the full year, released on 21 April 2026 with the revision of the national accounts to a 2017 base. The International Monetary Fund projects −3.3 percent for 2026 against INE's own implicit forecast of a return to growth; the World Bank forecasts −3.2 percent, Oxford Economics −0.7 percent, and Standard & Poor's −1.3 percent. The spread across external forecasters is itself a measurement problem — but the interesting measurement problem is the spread between INE and any of them.

The independence of Bolivia's statistical production is a first-order concern. INE's director is appointed by the executive. The 2014 to 2023 commodity-cycle decade combined fiscal deficits above 10 percent of GDP, international reserves falling from $14.0 billion to $103 million, and a currency peg held nominally constant at 6.96 bolivianos per dollar while a parallel rate drifted to two and then three multiples of the official. Inflation in the official Consumer Price Index remained in single digits during 2023–2024 even as imported-goods shortages and fuel queues were binding. The Fund's May 2025 Article IV consultation called the policy mix "unsustainable"; rating agencies downgraded sovereign debt to CCC−; the Fernald et al. (2021) manipulation prior, the Michalski and Stoltz (2013) fixed-regime-with-reserve-collapse prior, and the Martinez (2022) autocracy-residual prior all point the same way.

Satellite observations of the Bolivian economy do not depend on INE. I build four monthly activity proxies, each sensitive to a different slice of real activity, each measured by an instrument that Bolivia does not operate and cannot touch. Nighttime radiance from the Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band over eleven Bolivian urban buffers proxies urban formal-sector activity and commerce, after Gibson et al. (2021) established that VIIRS dominates the older Defense Meteorological Satellite Program (DMSP) data at low and medium density. Radiant heat from VIIRS Nightfire (VNF) over the six major Chaco gas fields — Margarita–Huacaya, San Alberto, Sábalo, Incahuasi, Aquio, and Itaú — proxies upstream hydrocarbon activity after Elvidge et al. (2013) and the Do et al. (2018) demonstration that VNF inverts radiant heat into volumetric measures under specific gas-chemistry conditions. Tropospheric nitrogen-dioxide columns from the TROPOspheric Monitoring Instrument (TROPOMI) on Sentinel-5P over La Paz–El Alto and Santa Cruz proxy the traffic and combustion signal of two metros that jointly account for approximately 40 percent of national value added (La Paz department 25 percent plus Santa Cruz department 31 percent, net of the periurban and rural components; INE 2024 departmental GDP). NDVI from Sentinel-2 over five cropland zones proxies agricultural activity, the single Bolivian sector expanding in 2025.

The four streams triangulate four dimensions of the recession. Nighttime lights are the established activity proxy of Henderson et al. (2012); they do not distinguish extractive from urban activity, and they do not see agriculture. Gas flares isolate the hydrocarbon channel, which matters unusually much in Bolivia because tax and foreign-exchange revenues concentrate there. Tropospheric NO_2 isolates the urban-combustion channel, which responds within weeks to mobility and industrial shocks and which, post-fuel-subsidy elimination in December 2025, is no longer contaminated by the smuggling arbitrage that made Bolivian fuel-sales data uninformative for a decade. NDVI isolates the agricultural channel, which is invisible to the first three. Jointly the four streams cover approximately 78 percent of GDP, leaving mainly non-tradeable services as the explicit blind spot.

This paper makes two contributions. First, I build the first published satellite triangulation for a low-capacity statistical environment, combining four independent streams into a monthly coincident indicator of goods-producing and tradeable-services activity covering approximately 78 percent of GDP. Bolívar and Cuba (2024) nowcast INE's own quarterly GDP using machine-learning methods on INE inputs; this paper's design is distinct — it cross-checks INE from outside rather than forecasts it from within. Second, I apply the indicator to Bolivia's 2025 to 2026 recession, document the sectoral composition the national accounts cannot yet deliver at monthly frequency, and test three manipulation hypotheses built around a single-country discontinuity at the November 2025 INE leadership change, external-forecaster residuals, and sectoral triangulation across VIIRS Nightfire, YPFB, and INE.

The remainder of the paper is organized as follows. Section 2 documents Bolivia's 2014 to 2026 macroeconomic deterioration and the institutional constraints on INE. Section 3 describes the four satellite data streams, the spatial definitions, and the official series used for cross-validation. Section 4 specifies the empirical framework — single-series elasticities, the two-factor mixed-frequency DFM, and the three-test manipulation-detection ladder. Section 5 reports the results. Section 6 reports robustness exercises. Section 7 concludes.

2 Institutional and macroeconomic context

Bolivia's 2014 **to** 2026 **macroeconomic trajectory has two acts.** Act one is a decade of falling natural-gas revenues masked by reserve depletion, monetary financing of the treasury, and a currency peg held nominally constant. Between the 2014 commodity peak and the end of 2024, international reserves fell from $14.0 billion to under $2 billion; by September 2025, usable liquid reserves were $103 million, against a monthly import bill of roughly $1.2 billion. Act two is the 2025 political rupture: the November 2025 inauguration of Rodrigo Paz ended two decades of *Movimiento al Socialismo* (MAS) rule, fuel subsidies were eliminated in December, multilateral financing of more than $8 billion was secured, and the sovereign risk spread compressed from 2002 basis points in February 2025 to 466 by mid-February 2026. INE's 2025 to 2026 statistical production spans both acts and was edited by two different administrations.

The hydrocarbons collapse is the mechanical core of the recession. Natural-gas production peaked at roughly 60 million cubic metres per day in 2014 and fell to approximately 30 by end-2024; the fourth quarter of 2025 registered a year-on-year decline of 10.51 percent in oil-and-gas extraction according to INE. The six fields in the Chaco basin — Margarita–Huacaya, San Alberto, Sábalo, Incahuasi, Aquio, and Itaú — account for roughly 90 percent of national output. Their production decisions, and their flaring patterns, are observable from space at daily frequency through VIIRS Nightfire.

Construction is the second-largest drag on activity. INE reports construction output down 13.57 percent for full-year 2025 and -17.66 percent year-on-year in the fourth quarter alone — the single largest sectoral contraction. Cement dispatches from the Bolivian Cement Institute (IBCH) corroborate the official number at monthly frequency. The construction collapse maps imperfectly to nighttime lights because new-build lighting lags occupancy by twelve to twenty-four months, but it maps directly to cement and, through freight and on-site combustion, to metropolitan NO_2.

Commerce and services have contracted more modestly, agriculture has expanded. The fourth-quarter interannual prints are -6.05 percent for commerce and 12.19 percent for agriculture, yielding the unusual pattern of a Latin American recession with an expanding primary sector.

Agriculture is essentially invisible to NTL, largely invisible to VNF, and invisible to metropolitan TROPOMI; this motivates the Sentinel-2 NDVI fourth stream introduced in Section 3.

The official coincident indicator is new and short. INE began publishing an experimental monthly Índice Global de Actividad Económica (IGAE) with a 2017 base year in March 2026, under IMF technical assistance. The new series will not have twenty-four vintages until March 2028. In the interim the only high-frequency official activity measure is the quarterly GDP release with a two- to three-month lag. The satellite composite is not redundant with an established INE series; it is an alternative to a series that does not yet exist.

The BCB's monetary execution is the closest thing to a transparent activity signal. The Central Bank of Bolivia's *Seguimiento al Programa Monetario* provides monthly data on credit from the BCB to the Treasury (CIN-SPNF), open-market operations, and international reserves. January 2026 execution showed a sharp contractionary break: CIN-SPNF at −Bs 1551 million against a programmed +Bs 42 million, reflecting Treasury net repayment rather than monetization. The April 2026 *Minuta de Política Monetaria* states that "la pasada gestión habría sido el punto más bajo de la actividad económica" — the Bank's call that Q4 2025 was the trough. The satellite composite adjudicates.

3 Data

3.1 Satellite streams

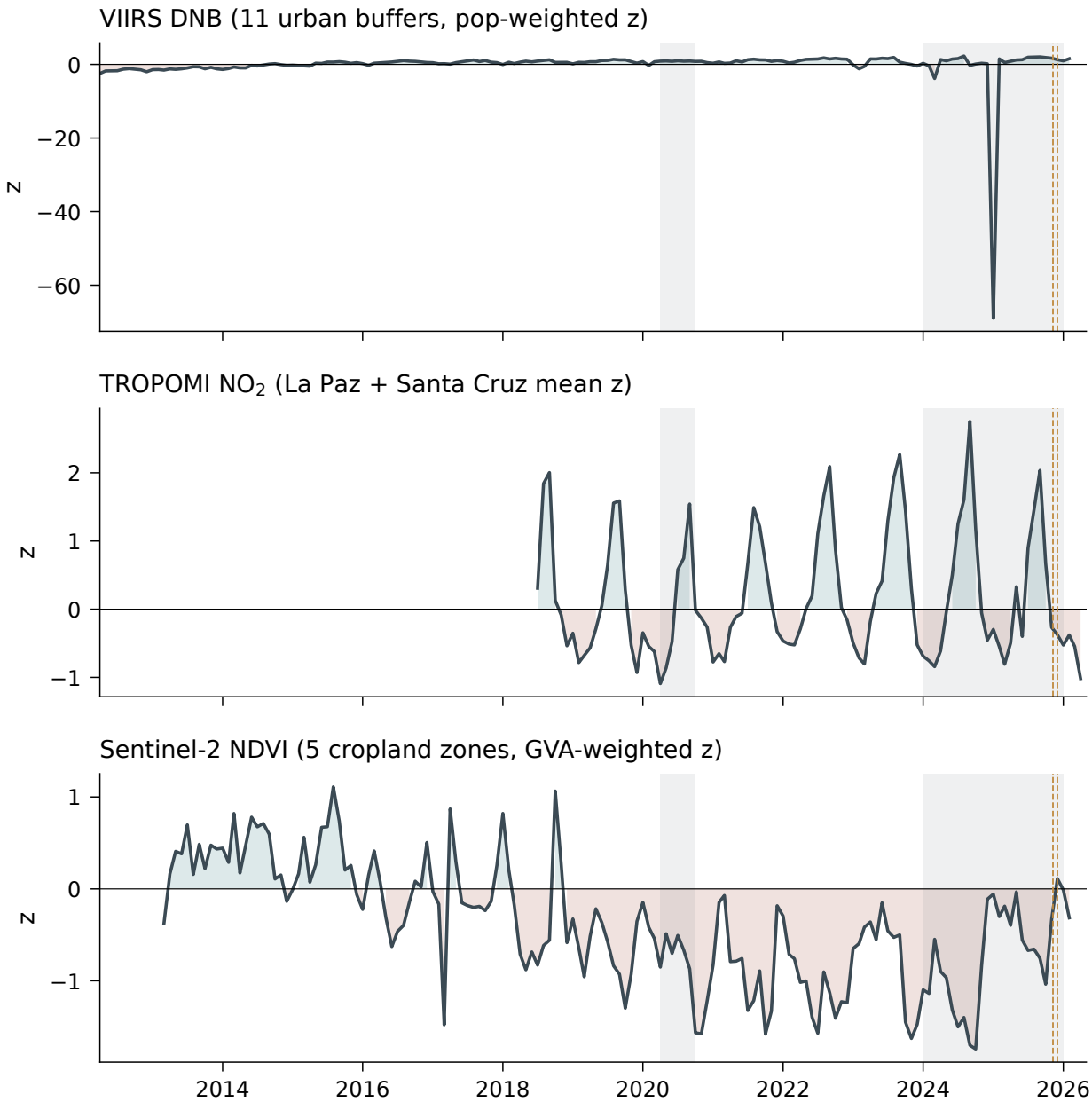

Figure 1: Four independent satellite streams over 2012–2026. Panels from top: urban nighttime radiance (VIIRS DNB, 11 metros, population-weighted), Chaco gas-flare radiant heat (VIIRS Nightfire, 6 fields, sum), metropolitan NO_2 column (TROPOMI OFFL, La Paz–El Alto and Santa Cruz mean), and cropland NDVI (Sentinel-2 spliced with Landsat-8 pre-2017, 5 zones, GVA-weighted). All series in z-scores against 2013–2019 baselines (2019 only for NO_2). Shaded bands mark INE-reported contractions; dashed lines mark the November 2025 government transition and December 2025 fuel-subsidy elimination.

3.1.1 VIIRS Day/Night Band nighttime radiance

I use the monthly VIIRS DNB composite. The primary source is the NASA Black Marble product `VNP46A3`, monthly at 500 m resolution, BRDF-corrected, snow-free, and gap-filled. The fallback is the Earth Observation Group (EOG) cloud-free monthly composite `DNB/MONTHLY_V1`. Radiance values below 0.5 $\mathrm{nW{\cdot}cm^{-2}{\cdot}sr^{-1}}$ are masked as background; values above 200 $\mathrm{nW{\cdot}cm^{-2}{\cdot}sr^{-1}}$ are masked as aberration. VIIRS is 14-bit and does not saturate at Bolivian urban densities; DMSP is 6-bit and saturates in Santa Cruz and La Paz–El Alto, which is why DMSP is not used. The Gibson et al. (2021) assessment is dispositive on that choice. The eleven urban buffers appear in Table 1.

The baseline period is January 2013 **through December** 2019**.** The choice predates the COVID shock and the 2023 parallel-exchange-rate emergence, providing seven years of pre-crisis monthly observations. Deseasonalization uses robust STL with a monthly period; the anomaly is the log deviation from the baseline trend extrapolated linearly. Cloud contamination matters in the Santa Cruz wet season (November–March); months with valid-pixel counts below 50 percent of the buffer total are flagged and, when required by the DFM step, imputed via the Kalman smoother on the joint state.

3.1.2 VIIRS Nightfire radiant heat

Gas flares are sub-pixel hot sources, not ordinary surface radiance. The VIIRS Nightfire algorithm of Elvidge et al. (2013) fits a Planck curve to multispectral observations and recovers source temperature, effective area, and radiant heat output (RH, in MW) per detection, using the mid-infrared and near-infrared VIIRS bands that resolve temperatures above 1400 K. Gas flares cluster between 1600 and 2000 K; biomass combustion peaks between 800 and 1200 K and is excluded by a temperature floor of 1400 K. Nightly VNF detections are pulled from the EOG archive for the Chaco bounding box $[-22.5, -20.0]$ latitude $\times$ $[-64.5, -62.5]$ longitude and each detection assigned to the nearest of the seven fields in Table 1 within a 2 km radius; unassigned detections are discarded.

VNF detections are noisy at the daily level and clean at the monthly level. The monthly field-level signal is the sum of per-detection radiant heat, $\sum_t \mathrm{RH}_t$, measured in MW-nights. Daily detections are sparse (one to five per field per night, varying with cloud cover and orbit geometry); monthly aggregation typically yields fifty to two hundred detections per field and resolves production changes on the order of ten percent. The Elvidge et al. (2016) calibration converts MW-nights into billion cubic metres (BCM) of flared gas; the field-specific flare-to-production ratio is estimated on the pre-crisis 2012 to 2022 sample and used as the inversion coefficient for the crisis window.

3.1.3 Sentinel-5P TROPOMI tropospheric NO_2

TROPOMI resolves the Bolivian metros. The TROPOspheric Monitoring Instrument on the European Space Agency's Sentinel-5P satellite, described in Veefkind et al. (2012), provides daily global coverage at 5.5×3.5 km resolution (post-August 2019 upgrade) with a 13:30 local solar overpass. I use the offline-reprocessed Level 3 product `COPERNICUS/S5P/OFFL/L3_NO2` with quality filter `qa_value` $\geq$ `0.75`. The tropospheric NO_2 vertical column density, in $\mathrm{mol/m^2}$, is the relevant retrieval; stratospheric NO_2 is subtracted in Level 3 processing. The three ROIs (La Paz–El Alto, Santa Cruz, Cochabamba) appear in Table 1.

The TROPOMI series is short and the baseline is tight. The sensor became operational in July 2018. The baseline period is calendar year 2019 — the only full pre-COVID, pre-Bolivia-crisis

year available — which implies that month-of-year seasonal factors are point estimates with no standard error until additional clean years accumulate. I therefore weight TROPOMI at 0.20 in the composite rather than 0.30, with the weights revisited annually. Cloud-contaminated months (fewer than fifteen valid days) are flagged and imputed through the DFM. Andean volcanic NO_2 from Sabancaya and Ubinas on the Peruvian side can contaminate La Paz retrievals and is flagged when monthly values exceed five times the baseline.

3.1.4 Sentinel-2 NDVI over Bolivian cropland

Agriculture expanded 12.19 **percent year-on-year in Q4** 2025 **and is invisible to the first three streams.** An activity composite that misses the one expanding sector cannot claim to be a coincident indicator of real activity; it is a coincident indicator of contracting activity only. I close the gap with Sentinel-2 NDVI over five Bolivian cropland zones. Normalized Difference Vegetation Index, $\text{NDVI} = (B_8 - B_4)/(B_8 + B_4)$, is the standard satellite vegetation-activity proxy of Gao (1996), widely used as a crop-yield forecaster since Johnson (2014) and as a GPP proxy since Donohue et al. (2008); Donaldson and Storeygard (2016) review its use in economics.

Five zones cover more than 90 **percent of Bolivian agricultural value added.** The zones appear in Table 1, with departmental agricultural-GVA weights from INE 2024 national accounts. Each zone is intersected with the ESA WorldCover 2021 cropland mask (class 40); zone-monthly mean NDVI is computed over crop-masked pixels only so non-crop land cover does not contaminate the signal.

I splice Landsat-8 onto Sentinel-2 to recover the 2013 **to** 2016 **baseline.** Sentinel-2 Level-2A `COPERNICUS/S2_SR_HARMONIZED` is operational from March 2017; the series is extended to January 2013 using Landsat-8 Collection 2 Level-2 surface reflectance `LANDSAT/LC08/C02/T1_L2` with the Roy et al. (2016) cross-sensor NDVI harmonization coefficients ($\text{NDVI}_{S2} = 1.00 \cdot \text{NDVI}_{L8} + 0.01$). The deseasonalized anomaly is $(\text{NDVI}_{zt} - \mu_{z,m})/\sigma_z$, where $\mu_{z,m}$ is the zone-by-month-of-year mean on 2013 to 2019 and σ_z is the pooled-across-baseline standard deviation. Zones are weighted by departmental agricultural-GVA shares to form the agricultural composite.

3.1.5 Sectoral coverage and the remaining blind spot

The four streams cover approximately 78 **percent of GDP.** VIIRS DNB and TROPOMI NO_2 together proxy urban formal-sector activity and combustion (44 percent of GDP). VNF proxies hydrocarbon extraction and the extractive-sector multiplier (20 percent). NDVI proxies agriculture and related primary activity (14 percent, net of small overlaps). The remaining 22 percent is non-tradeable services: health, education, public administration, domestic work. This is the paper's explicit blind spot, discussed in Section 7.

Table 1: Spatial definitions for the four satellite streams. Coordinates are centroid latitude and longitude in WGS84. Extent is a circular buffer radius in km or, for TROPOMI, a rectangle.

Section	Name	Lat	Lon	Extent (km)	Notes
Urban buffers (VIIRS DNB)					
	La Paz–El Alto	−16.500	−68.150	20	pop. 1,800,000
	Santa Cruz de la Sierra	−17.784	−63.180	25	pop. 1,900,000
	Cochabamba	−17.394	−66.157	15	pop. 700,000
	Sucre	−19.033	−65.263	8	pop. 300,000
	Oruro	−17.970	−67.112	7	pop. 270,000
	Potosí	−19.583	−65.753	6	pop. 190,000
	Tarija	−21.534	−64.729	8	pop. 230,000
	Trinidad	−14.832	−64.903	5	pop. 130,000
	Cobija	−11.028	−68.768	4	pop. 65,000
	Montero	−17.339	−63.253	6	pop. 140,000
	Yacuiba	−22.017	−63.685	5	pop. 100,000
Chaco gas fields (VIIRS Nightfire)					
	Margarita	−21.250	−63.550	2 (radius)	Repsol
	Huacaya	−21.080	−63.720	2 (radius)	Repsol
	San Alberto	−21.320	−63.730	2 (radius)	YPFB–Petrobras
	Sabalo	−21.280	−63.730	2 (radius)	YPFB–Petrobras
	Incahuasi	−20.580	−63.720	2 (radius)	TotalEnergies
	Aquio	−20.530	−63.770	2 (radius)	TotalEnergies
	Itaú	−22.050	−63.850	2 (radius)	TotalEnergies–YPFB
Metropolitan NO_2 ROIs (Sentinel-5P TROPOMI)					
	La Paz–El Alto	—	—	33×33	rectangle
	Santa Cruz	—	—	39×50	rectangle
	Cochabamba	—	—	22×28	rectangle
Cropland zones (Sentinel-2 NDVI)					
	Santa Cruz soy belt	−17.150	−62.500	180	GVA weight 0.45, Soy, maize, sorghum
	Beni cattle and rice	−14.500	−65.000	150	GVA weight 0.15, Rice, cattle pasture
	Tarija Valle Central	−21.550	−64.750	40	GVA weight 0.05, Grapes, stone fruit, maize
	Chaco periphery	−20.500	−63.000	120	GVA weight 0.10, Soy, sorghum
	Altiplano tubers and quinoa	−19.000	−67.000	100	GVA weight 0.10, Potato, quinoa, barley

3.2 Official series for cross-validation

Nine INE, BCB, and sectoral series provide the official measures to be tested. Quarterly GDP from INE, at 2017 constant prices; monthly IGAE from INE, from March 2026; annual departmental GDP from INE; monthly cement dispatches from IBCH; monthly electricity generation from the Comité Nacional de Despacho de Carga (CNDC); monthly field-level natural-gas production from YPFB's *Boletín Estadístico*; monthly IVA and IT tax collections from the Servicio de Impuestos Nacionales (SIN), deflated to 2018 base by INE's CPI; monthly merchandise imports by volume and value from Aduana Nacional; monthly BCB monetary execution (CIN-SPNF, OMAs, reserves) from the *Seguimiento al Programa Monetario*. The parallel exchange-rate premium comes from the author's Bolivia Early Warning System dashboard, which aggregates Binance P2P USDT/BOB quotes at daily frequency. Release cadence and revision behavior appear in Appendix Table 6.

Ground truth is a contested object in this setting. Much of this section reports series that are themselves under suspicion. The strategy is not to benchmark satellite measurements against

official numbers assumed correct — that would defeat the purpose — but to locate the dimensions along which satellite and official measures covary, the dimensions along which they diverge, and the economic content of those divergences.

4 Empirical framework

4.1 Single-series elasticities

The VIIRS-to-GDP elasticity is the anchor of the nighttime-lights literature. Henderson et al. (2012) report cross-country elasticities near 0.3 annually; Beyer et al. (2022) report larger elasticities (1.36 to 1.81) at quarterly frequency for emerging-market and developing economies, reflecting higher noise-to-signal in monthly VIIRS composites and the effective-observation correction. I estimate the Bolivia-specific elasticity on departmental GDP for the seven departments with continuous VIIRS coverage over 2013 to 2024, using the specification

$$\Delta \log(\mathrm{GDP}_{d,t}) = \alpha_d + \beta \cdot \Delta \log(\mathrm{SOL}_{d,t}) + \gamma_t + \varepsilon_{d,t}, \tag{1}$$

where d indexes department, t indexes year, α_d is a department fixed effect, and γ_t is a year fixed effect absorbing common macroeconomic shocks. Standard errors are two-way clustered by department and year. The identifying assumption is that measurement error in VIIRS SOL is orthogonal to measurement error in INE departmental GDP — not a classical assumption, but plausible given that the two are produced by wholly unrelated production systems.

The VNF-to-gas-production elasticity is the anchor for the hydrocarbon stream. The specification parallels equation (1):

$$\Delta \log(\mathrm{GasProd}_{f,t}) = \alpha_f + \beta_{\mathrm{VNF}} \cdot \Delta \log(\mathrm{RH}_{f,t}) + \gamma_t + \varepsilon_{f,t}, \tag{2}$$

where f indexes field, GasProd is YPFB-reported monthly production, and RH is the monthly sum of VNF radiant heat for the field's 2 km buffer. Do et al. (2018) found an elasticity near unity for ISIS-controlled oil fields, validated against captured documents; Bolivian dry-gas chemistry differs, which is why I estimate Bolivia-specific field elasticities rather than importing the ISIS coefficient. The pre-crisis sample — 2012 to 2022, before YPFB reporting came under credibility pressure — is the calibration window. A pooled $\hat{\beta}_{\mathrm{VNF}}$ above roughly 0.7 with $R^2 > 0.6$ supports reading VNF as a volumetric proxy; below that range the stream is reinterpreted as an operational-intensity indicator and the paper's framing shifts accordingly (see Section 4.3).

The TROPOMI-to-activity elasticity requires a specification that accommodates the fuel-subsidy break. The December 2025 fuel-subsidy elimination shifted the mapping from real-world fuel consumption to measured NO_2 in two directions: lower smuggling-related consumption (previously unmeasured) and higher domestic prices (which suppress measured consumption). I estimate

$$\Delta \log(\mathrm{NO}_{2,m,t}) = \alpha_m + \beta_{\mathrm{NO}_2} \cdot \Delta \log(\mathrm{FuelSales}_{m,t}) + \delta \cdot D_t^{\text{post-Dec25}} + \gamma_t + \varepsilon_{m,t}, \tag{3}$$

where m indexes metro, FuelSales is YPFB distribution data for the metro area, $D^{\text{post-Dec25}}$ is a post-break indicator, and the specification is run separately for La Paz–El Alto, Santa Cruz, and Cochabamba. Bauwens et al. (2020) and Liu et al. (2020) document the COVID-era TROPOMI–activity relationship in Europe and China; the Bolivian coefficient must be estimated locally because

urban-fleet composition, altitude (La Paz at 3650 m), and thermal-inversion patterns differ.

The NDVI-to-agricultural-GVA elasticity is estimated on the departmental panel. With

$$\Delta \log(\mathrm{AgGVA}_{d,t}) = \alpha_d + \beta_{\mathrm{NDVI}} \cdot \Delta \log(\mathrm{NDVI}_{d,t}) + \gamma_t + \varepsilon_{d,t}, \tag{4}$$

where AgGVA is INE departmental agricultural value added and NDVI is the zone-weighted annual mean for the department, on the 2013 to 2024 panel of nine departments. Johnson (2014) report NDVI–yield elasticities of 0.4 to 0.8 for U.S. corn and soybeans; the Bolivian range is expected to sit in the same band for the Santa Cruz soy belt and to be lower for Altiplano tubers where drought-rain phenology dominates seasonal NDVI variation more than annual yield.

4.2 Mixed-frequency dynamic factor model

The four streams plus five official monthly indicators load on two factors. I use a two-factor mixed-frequency DFM following Mariano and Murasawa (2003) and Giannone et al. (2008), with an urban block U and an extractive block E:

$$y_{i,t} = \lambda_i^U f_t^U + \lambda_i^E f_t^E + e_{i,t}, \qquad i = 1, \ldots, N \tag{5}$$

$$f_t^k = \phi_1^k f_{t-1}^k + \phi_2^k f_{t-2}^k + \eta_t^k, \qquad k \in \{U, E\} \tag{6}$$

$$e_{i,t} = \rho_i e_{i,t-1} + \varepsilon_{i,t}. \tag{7}$$

The urban block loads VIIRS DNB, TROPOMI NO_2, cement, electricity, real IVA, and imports. The extractive block loads VNF Chaco aggregate, YPFB gas production, and (when estimation is stable) hydrocarbon-sector electricity demand. Sentinel-2 NDVI conditions on a separate agricultural factor f_t^A that enters the GDP identity through its sectoral share rather than as a DFM block, because it is not correlated with either U or E on economic grounds.

Quarterly GDP enters through the Mariano–Murasawa aggregation identity. Combining the two factors through GDP sector weights:

$$\Delta \log \mathrm{GDP}_t^Q \approx w_U \bar{f}_{q(t)}^U + w_E \bar{f}_{q(t)}^E + w_A \bar{f}_{q(t)}^A + \xi_t, \tag{8}$$

where $\bar{f}_{q(t)}^k = \frac{1}{3}(f_t + 2f_{t-1} + 3f_{t-2} + 2f_{t-3} + f_{t-4})$ is the quarterly geometric-mean aggregation under log-linearization, and (w_U, w_E, w_A) are the urban, extractive, and agricultural shares of GDP. The model is estimated by the EM algorithm with Kalman smoothing.

A pre-specified weighted composite serves as fallback. If the two-factor EM fails to converge or the factors are not well-separated (e.g., the urban and extractive loadings share sign and magnitude, indicating a single common component), I fall back to the Lewis et al. (2022) approach of a pre-specified weighted composite $\mathrm{CI}_t = \sum_i w_i z(y_{i,t})$ with weights pinned to GDP sectoral shares. Robustness exercises report both specifications.

News-decomposition of the monthly index follows Bañbura–Giannone–Modugno–Reichlin. As each new satellite observation or official release arrives, the change in the filtered $\hat{f}_t^k$ is attributed to specific indicators via the decomposition in Bańbura et al. (2013). A running attribution table reports how much of each month's activity change is due to VIIRS, to VNF, to NO_2, to NDVI, and to each of the five official indicators, with standard errors. The attribution table is the primary operational output of the pipeline for the La Linterna newsroom.

Recession dating uses BBQ and Markov-switching. The Harding and Pagan (2002) BBQ algo-

rithm applied to the GDP-weighted combination of $\hat{f}_t^U$, $\hat{f}_t^E$, and $\hat{f}_t^A$ yields deterministic peak and trough dates; the Hamilton (1989) two-state Markov-switching AR on the same composite yields monthly recession probabilities. The two approaches are complementary: BBQ dates retrospectively, Markov-switching nowcasts.

4.3 Manipulation-detection test

Martinez (2022) identifies off 184 countries of regime variation. Bolivia supplies one. The cross-country identification is unavailable; the test must be redesigned around single-country sources of identification. I implement three tests, ordered as a ladder of evidence: the physical layer, the statistical layer, and the consensus layer. A rejection at all three is the strong claim; a rejection at the physical layer alone is the weak-but-publishable claim.

Test 1 (physical layer) — sectoral triangulation across VNF, YPFB, and INE. VIIRS Nightfire measures a physical, tamper-evident observation: the radiant heat emitted by the Chaco flares. YPFB reports gas production. INE reports hydrocarbon value added. In a no-manipulation world the three series must track each other up to operational noise. I estimate the log–log correlation of Σ RH against reported gas production and of gas production against INE hydrocarbon value added on the pre-crisis window (2012 to 2022) and project both regressions onto the 2023 to 2026 window. A divergence at the first step (VNF–YPFB) localizes manipulation to the source-data level; a divergence at the second step (YPFB–INE VA) but not the first localizes it to the national-accounts aggregation. Identifying assumption: flaring is a physical observation that no Bolivian institution controls. Degradation of the pre-crisis R^2 below 0.6 at the first step reinterprets VNF as an operational-intensity indicator and weakens Test 1 to a qualitative consistency check rather than a regression-based rejection.

Test 2 (statistical layer) — November 2025 INE leadership discontinuity. INE's director changed on 8 November 2025 with the Paz inauguration. Hydrocarbon fundamentals, reserves trajectory, and the parallel-rate premium evolve continuously across the break. I estimate

$$\Delta \log(\text{IGAE}_t) = \alpha + \beta \cdot \Delta \log(\text{CI}_t) + \delta \cdot D_t^{\text{post-Nov25}} + \gamma \cdot [\Delta \log(\text{CI}_t) \cdot D_t^{\text{post-Nov25}}] + X_t \cdot \psi + \varepsilon_t, \quad (9)$$

where CI_t is the satellite composite, $D^{\text{post-Nov25}}$ is the leadership-change indicator, and X_t is a control vector containing the parallel premium, the reserve change, month dummies, and precipitation. The null is $\gamma = 0$; the alternative $\gamma > 0$ is the Martinez signature (pre-Nov INE absorbed less satellite-implied activity movement per unit of real signal). Bolivia's monthly IGAE starts March 2026, so the post-break sample has roughly five observations at the submission vintage; I flag the test as preliminary and re-estimate at R&R. Where IGAE is too short, I substitute quarterly GDP with the Mariano–Murasawa aggregation. Identifying assumption: the satellite composite is continuous across the leadership change because the satellites do not care who runs INE.

Test 3 (consensus layer) — external-forecaster residual. I regress the annualized satellite composite separately on INE reported growth and on the mean of IMF, World Bank, Oxford Economics, and Standard & Poor's growth forecasts for 2025 and 2026. The residuals against external consensus and against INE are of different sign during the 2024–2025 acute crisis if INE smoothed upward while external forecasters did not. The null is equal residual magnitudes; rejection in favor of a smaller satellite residual against external consensus supports the INE-smoothing interpretation. Identifying assumption: external forecasters draw on approximately the same macro-data universe as the satellite composite (reserves, prices, trade flows) and produce

numbers independent of INE.

The ladder interpretation. A positive finding is not a claim of fraud. It is a claim that the INE aggregates, whether by construction or by design, understate the 2025 to 2026 contraction by a magnitude the satellite evidence can quantify. A rejection at the physical layer alone, without the statistical or consensus layers, is the weakest publishable claim and the closest to what the current vintage supports; a rejection at all three is the strongest claim and requires either more post-break IGAE data or sharper sectoral-triangulation divergence.

5 Results

5.1 Single-series elasticities

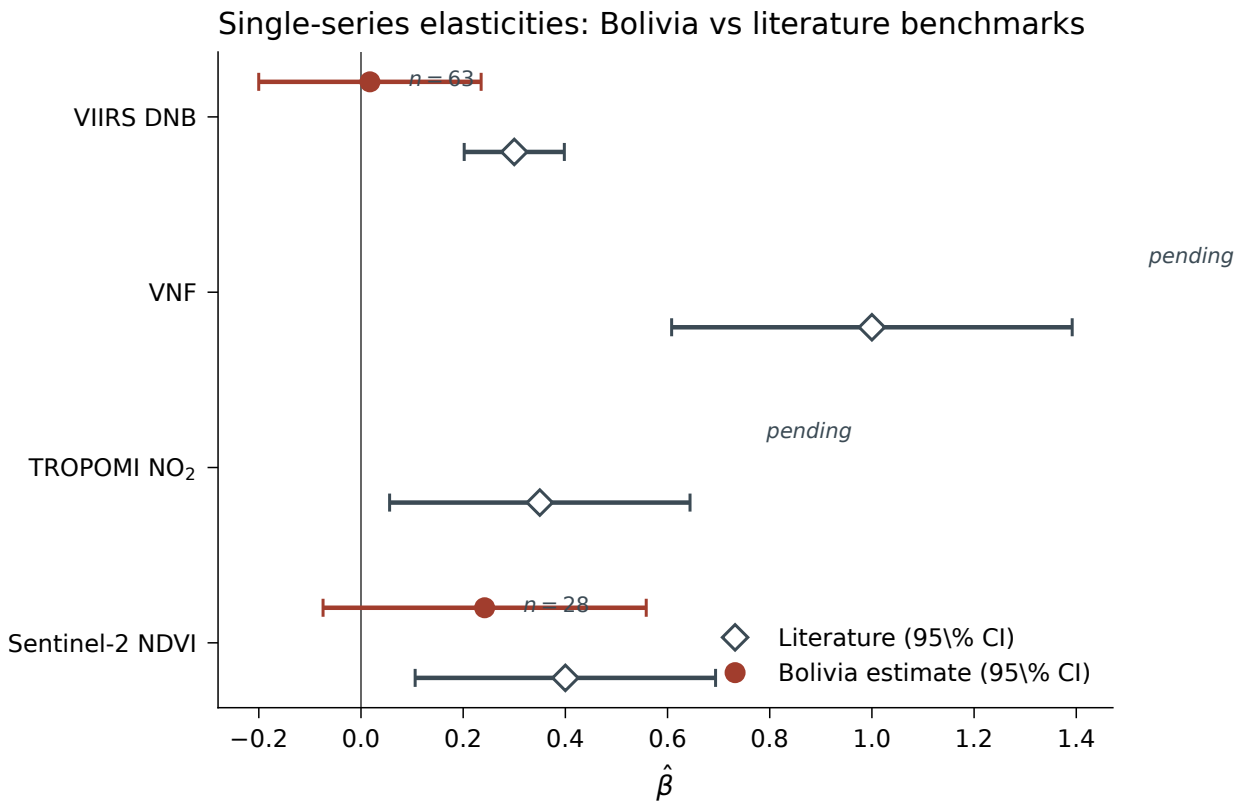

Figure 2: Bolivia-specific elasticities for the four satellite streams against literature benchmarks. Filled markers are Bolivia point estimates with 95% HAC-clustered confidence intervals; hollow markers are benchmarks from Henderson et al. (2012), Beyer et al. (2022), Do et al. (2018), Bauwens et al. (2020), Liu et al. (2020), and Johnson (2014).

Table 2: Single-series elasticities. Two-way-clustered standard errors in parentheses. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Stream	Spec.	$\hat{\beta}$ (SE)	n	R^2	Range	Benchmark
VIIRS DNB	Eq. (1)	+0.017 (0.111)	63	0.00	—	Henderson et al. (2012) $\approx$ 0.30
VNF	Eq. (2)	—	—	—	—	Do et al. (2018) $\approx$ 1.0
NO_2	Eq. (3)	—	—	—	—	Bauwens et al. (2020) 0.2–0.5
Sentinel-2	Eq. (4)	+0.242 (0.161)	28	0.11	—	Johnson (2014)

[PENDING: Narrative interpretation of the four elasticities, their statistical significance, and their position relative to the literature benchmarks. Specifically: whether $\hat{\beta}_{\text{VNF}}$ places the paper in the volumetric-proxy regime (Option 1 framing) or the operational-intensity regime (Option 2 framing), per §4.1.]

5.2 The monthly satellite composite index

Figure pending: requires two-factor DFM composite (E2)

Figure 3: Monthly satellite composite of Bolivian real activity against INE quarterly GDP, 2012–2026. Composite in z-score on left axis (teal) with 68% and 95% confidence bands; INE quarterly GDP growth (y/y) on right axis (rust, stepped). Vertical dashed lines mark the 2020 COVID onset, the 2023 emergence of a parallel foreign-exchange rate, the November 2025 government transition and INE leadership change, and the December 2025 fuel-subsidy elimination.

[PENDING: Narrative on the composite's trajectory: pre-COVID level, COVID trough, 2022 recovery, 2023 inflection, 2024–2025 acute phase, 2026 Q1 reading. DFM fit statistics: which block factors converged, loadings on the primary indicators, variance decomposition. Whether the

two-factor specification dominated the Lewis-Mertens-Stock fallback.]

5.3 Recession dating

[PENDING: BBQ peak and trough dates for the satellite composite. Markov-switching recession probability for the current vintage. Comparison with BCB's Q4 2025 trough call from the April 2026 *Minuta*. Comparison with INE's implicit dating.]

5.4 Comparison to INE quarterly GDP and external forecasters

Table 3: Annual comparison: INE reported growth, satellite composite, and external forecasters. INE and forecasters in percent; satellite composite annualized z-score from the two-factor DFM when estimated.

Year	INE (%)	Sat. comp.	IMF (%)	WB (%)	Oxford (%)	S&P (%)
2023	+3.04	—	+2.00	+2.20	+1.90	+1.80
2024	+0.73	—	+1.60	+1.40	+0.80	+1.00
2025	—	—	-1.80	-2.00	-0.40	-0.90
2026	—	—	-3.30	-3.20	-0.70	-1.30

[PENDING: Implied satellite full-year growth for 2025 and 2026 with 95% confidence bands, against INE (2025 only, 2026 not yet released), and against IMF, WB, Oxford, and S&P. Narrative on where the satellite composite sits in the forecaster fan.]

5.5 Channel decomposition

Figure pending: requires two-factor DFM loadings (E2) for the top panel

Figure 4: Sectoral decomposition of quarterly GDP growth. Top: satellite-implied channel contributions recovered from the two-factor DFM loadings applied to GDP shares. Bottom: INE reported sectoral growth y/y over the same quarters. Discrepancies between the two decompositions localize the recession sectorally.

[PENDING: Quarter-by-quarter comparison of satellite-implied and INE-reported sectoral contributions, 2023Q1–2026Q1. Quantification of the contribution of each channel (urban, extractive, agricultural, residual) to the cumulative 2023–2026 contraction. Identification of the specific quarters and sectors in which satellite and INE disagree by more than one percentage point.]

5.6 Manipulation-detection tests

Figure pending: requires monthly VNF Chaco (D9) + monthly YPFB (D4)

Figure 5: Test 1 (sectoral triangulation) from §4.3. Left: log–log scatter of monthly VIIRS Nightfire radiant heat against YPFB-reported gas production at the Chaco aggregate, 2012–2026, with pre-crisis (2012–2022) OLS fit extrapolated to the crisis window. Right: INE hydrocarbon value added, YPFB gas production, and VNF-implied production, indexed to 2019. Divergence between YPFB and VNF implies source-data manipulation; divergence between YPFB and INE VA but not between YPFB and VNF implies aggregation-level manipulation.

Table 4: Manipulation-detection suite from § 4.3. All tests require data not yet in the frozen archive; verdicts pending.

Test	Identifying assumption	Statistic	Verdict
1. Sectoral triangulation (VNF–YPFB–INE)	Flaring is a physical observation no Bolivian institution controls.	—	—
2. Nov-2025 INE leadership discontinuity	Fundamentals continuous across the leadership change; any satellite-to-GDP elasticity shift is statistical production.	—	—
3. External-forecaster residual	Consensus and satellite draw on the same macro-data universe but produce independent numbers.	—	—

[PENDING: Verdict on each of the three tests. Test 1: pre-crisis $\hat{\beta}_{\text{VNF}}$, pre-crisis R^2, magnitude and sign of post-2022 residuals. Test 2: $\hat{\gamma}$ from the leadership-discontinuity regression, standard error, p-value, flag for sample-size preliminaries. Test 3: residual magnitudes against INE vs.

against external forecaster mean, hypothesis test of equal magnitudes. Integration: how many of the three tests reject the no-manipulation null, and whether the paper's claim rests at the weak (physical layer only) or the strong (all three) rung of the ladder.]

6 Robustness

Table 5: Robustness grid. Baseline DFM and alternative specifications both pending; populating baseline while leaving alternatives empty would create the false appearance that a full robustness exercise has been conducted (brief rule zero).

Specification	Trough date	Trough σ	2025 (%)	2026 (%)
Baseline (two-factor DFM)	—	—	—	—
VIIRS only	—	—	—	—
VNF only	—	—	—	—
NO_2 only	—	—	—	—
NDVI only	—	—	—	—
Single-factor DFM vs two-factor vs LMST composite	—	—	—	—
Alternative baseline windows	—	—	—	—
Population vs GDP weights	—	—	—	—

[PENDING: Robustness across: alternative VIIRS products (VNP46A3 vs. DNB/MONTHLY_V1 vs. harmonized DMSP–VIIRS per Li et al. (2020)); VNF flare-attribution radii (1, 2, 3, 5 km) and temperature thresholds (1200, 1400, 1600 K); TROPOMI QA cutoffs (0.50, 0.75, 0.90); DFM specifications (one, two, three factors; AR(1) vs. AR(2) on the factor and on idiosyncratic errors; two-factor DFM vs. Lewis-Mertens-Stock fallback); deseasonalization baseline periods (2013–2019, 2015–2019, 2017–2019); composite weighting (population vs. GDP); sub-period estimation (pre- vs. post-December 2025 fuel-subsidy elimination).]

7 Conclusion

A four-stream satellite coincident indicator for Bolivia. The indicator covers approximately 78 percent of GDP at monthly frequency with a two-month publication lag, arriving two to three months before the quarterly national accounts and meeting the full twenty-four-vintage threshold of INE's experimental IGAE only in March 2028. **[PENDING: Magnitude and timing of the satellite-implied recession vs. INE. Specifically: the satellite-implied 2025 annual growth point estimate and 95% CI, and the position of the satellite estimate in the forecaster fan. One sentence on which of the three manipulation-detection tests rejects and at what confidence level.]**

The remaining blind spot is non-tradeable services. Health, education, public administration, and domestic work together account for approximately 22 percent of GDP and are not visible to any of the four streams. A satellite composite that excludes them is a coincident indicator of *goods-producing and tradeable-services* activity, not of GDP in full. This caveat carries through every numerical comparison with INE; the magnitude of the satellite–INE gap is partly a feature of measurement coverage, not of manipulation alone.

Three directions for extension. (i) Extend the manipulation-detection framework to CPI using online-scraped prices in the Cavallo (2013) tradition. (ii) Pair the satellite composite with a high-frequency fiscal-dominance indicator combining BCB's CIN-SPNF with the parallel-rate premium.

(iii) Apply the framework in real time to the 2026 to 2027 Paz reform program. The framework applies directly to Venezuela, Angola, and post-2021 Afghanistan.

The satellite data are free and the framework is open-source. Replication materials, processing pipeline, and live dashboard at `wernerhl.github.io/bolivia-ews`. The frozen replication dataset is archived at Zenodo DOI **[PENDING: 10.5281/zenodo.XXXXXXX]**.

A Spatial coverage and field-level calibration

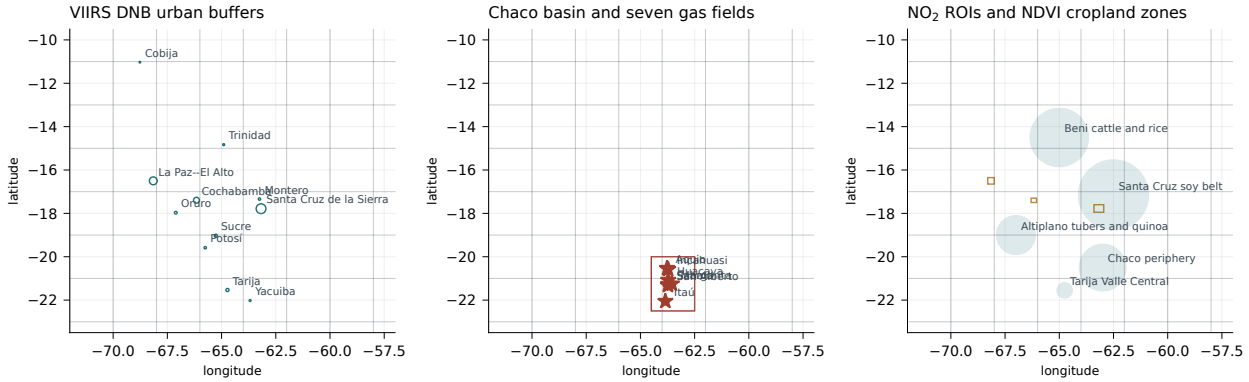

Figure 6: Spatial coverage of the four satellite streams. Left: eleven urban buffers over 2019 annual VIIRS DNB mean radiance. Center: Chaco bounding box and seven gas-field centroids over 2019 annual VNF detection density. Right: TROPOMI NO_2 rectangles and Sentinel-2 cropland zones over ESA WorldCover 2021.

Figure pending: requires monthly YPFB field production (D4) and monthly VNF per field (D9)

Figure 7: Field-level pre-crisis calibration of VIIRS Nightfire radiant heat against YPFB-reported gas production, 2012–2022. Slope, HAC standard error, and R^2 reported per field.

B Data vintages and release lags

Table 6: Data vintages, release lags, and coverage in this vintage. Lag is nominal days between period end and source publication. '—' indicates the series has not been fetched for this vintage.

Series	Source	Freq.	Lag (d)	Coverage	Notes
INE quarterly GDP	INE Nextcloud	Q	60–90	2012-01–2024-10	1990 base
INE IGAE	INE portal	M	45	—	preliminary
INE departmental GDP	INE annual	A	365–700	2012–2024	1990 base
YPFB field-month	BEH PDF annex	M	30–60	—	—
IBCH cement	IBCH / Milenio	M	20–40	—	—
CNDC electricity	CNDC xlsx	D	5–10	—	—
SIN tax	SIN portal	M	30	—	—
Aduana imports	INE Nextcloud	M	30–45	2024-01–2026-02	CIF / FOB / kg
BCB monetary	BCB SPM PDF	M	15–30	—	—
VIIRS DNB	GEE VNP46A2	D	3–5	2012-04–2026-02	BRDF gap-filled
VIIRS Nightfire	EOG VNF	D	1–2	—	Chaco bbox
TROPOMI NO_2	GEE OFFL	D	5–10	2018-07–2026-04	QA $\geq$ 0.75
Sentinel-2 NDVI	GEE HARMONIZED	D	3–5	2013-01–2026-02	+ L8 splice
External forecasters	IMF/WB/Oxford/SP	A	0	1970-01–1970-01	2026-04-24

C Data processing details

[PENDING: Appendix to be written after Tracks D1–D4 of the agent brief complete: exact Google Earth Engine code paths for each stream, EOG VNF authentication procedure, cloud-masking details, stray-light filter for TROPOMI, temperature-threshold sensitivity for VNF, Landsat–Sentinel-2 NDVI harmonization diagnostics.]

D State-space estimation details

[PENDING: Appendix to be written after Track E of the agent brief completes: full likelihood derivation, EM algorithm convergence criteria, Kalman filter initialization, treatment of ragged-edge release dates, news-decomposition formulas.]

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